From Electronic Structure to Continuum Response: A Computational Model for Strain-Dependent Conductivity in Piezoresistive Materials

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Abstract

Many sensing devices rely on transduction mechanisms that convert mechanical deformation into measurable electrical signals. A prominent example is the piezoresistive effect, where the electrical conductivity of a material varies under applied strain.

From a materials physics perspective, this behavior originates from strain-induced modifications of the electronic structure and carrier transport properties. Mechanical deformation alters band energies, effective masses, and scattering mechanisms, which in turn influence carrier mobility and electrical conductivity [3]. In certain regimes, additional effects such as tunneling may also contribute to the overall response.

This work develops a computational framework for modeling strain-dependent conductivity in deformable materials by connecting these physical mechanisms across scales. At the continuum level, the mechanical behavior is described by the equations of elasticity, while the electrical response is governed by a conductivity equation with strain-dependent coefficients. This leads to a coupled system of partial differential equations of the form

$$\nabla \cdot \boldsymbol{\sigma}(u) = 0, \qquad \nabla \cdot (\sigma(\varepsilon(u))\nabla V) = 0,$$

where u denotes the displacement field, V the electrical potential, and $\sigma(\varepsilon(u))$ represents an effective conductivity incorporating the influence of mechanical deformation.

A central aspect of this study is the interpretation of the constitutive relation $\sigma(\varepsilon)$. Rather than treating it purely phenomenologically, the framework is designed so that it can be informed by transport models derived from electronic structure calculations, providing a physically motivated link between microscopic behavior and macroscopic response.

From a computational perspective, the focus lies on the formulation and numerical solution of the coupled system using finite element methods. A baseline case with constant conductivity is first analyzed and validated, followed by the extension to spatially varying, strain-dependent conductivity. The framework is constructed to remain flexible, allowing the incorporation of physics-based models or data-driven approximations for $\sigma(\varepsilon)$.

The proposed approach provides a consistent and extensible pathway for studying electromechanical coupling in materials. Potential applications include sensing technologies, flexible electronics, and structural health monitoring.

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1 Introduction

The piezoresistive effect arises from the coupling between mechanical deformation and electrical conduction. In typical applications, a doped semiconductor element (such as silicon) is embedded within a mechanical structure, where external forces induce strain in the material. This deformation alters the electronic transport properties of the material, leading to measurable changes in its electrical response, commonly observed as variations in voltage or resistance.

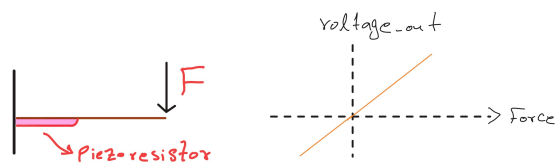

Figure 1: A schematic representation of the piezoresistive effect, where mechanical deformation leads to changes in electrical response.

The piezoresistive effect, first observed by William Thomson (Lord Kelvin) in the 19th century and later developed for semiconductors by Bardeen, Shockley, and Smith [4], describes the dependence of electrical conductivity on mechanical deformation. At a microscopic level, this behavior originates from strain-induced modifications of the electronic structure and carrier transport mechanisms. In particular, quantities such as band energies, effective masses, and scattering processes are affected by deformation, which in turn influence carrier mobility and conductivity [3].

There exists a wide spectrum of engineering realizations of the piezoresistive effect, including strain gauges, pressure sensors, microelectromechanical systems (MEMS), cantilever-based sensing devices, and flexible or wearable systems [1]. Although these systems differ in geometry, material composition, and boundary conditions, they can be interpreted as different realizations of the same underlying electromechanical coupling.

Despite extensive technological development and optimization, the modeling of this coupling is often carried out in a phenomenological and application-specific manner, where the dependence of conductivity on strain is introduced empirically. From a computational materials perspective, this raises a more fundamental question: how can such macroscopic descriptions be related in a consistent way to the underlying physical mechanisms governing electronic transport?

The present work approaches this problem by developing a computational framework in which the electromechanical coupling is described at the continuum level, while maintaining a clear connection to the underlying transport physics. In this setting, the continuum model serves as an

effective description of strain-dependent conductivity, acting as a bridge between microscopic electronic behavior and macroscopic response.

2 Problem Articulation

At the continuum level, electrical conduction in a deformable material can be described by a conductivity equation in which the electrical conductivity depends on the local mechanical strain. Such models are widely used in engineering applications and can be solved efficiently using numerical methods. However, the central quantity in this description, namely the strain-dependent conductivity $\sigma(\varepsilon)$, is often introduced phenomenologically, without an explicit connection to the underlying material physics.

From a materials physics perspective, this raises a fundamental question: how does mechanical deformation at the atomic scale translate into a change in macroscopic electrical conductivity?

In semiconductors, strain modifies the crystal structure and therefore the electronic band structure of the material. This leads to changes in quantities such as band energies, effective masses, and carrier velocities. As a consequence, the transport properties of charge carriers are altered, affecting mobility and electrical conductivity. These mechanisms are described in semiconductor theory through band-structure analysis and transport models such as the semiclassical Boltzmann transport equation [3].

This observation suggests a natural multiscale structure of the problem. At the microscopic level, strain influences the electronic structure of the material. At the transport level, the modified electronic structure determines carrier dynamics and effective conductivity. At the continuum level, this conductivity enters as a constitutive parameter in the electrical conduction equation.

The core challenge of this work is therefore to connect these levels in a computationally consistent way. Rather than treating $\sigma(\varepsilon)$ purely as a fitting function, the goal is to interpret it as an effective quantity that reflects underlying electronic and transport mechanisms, while remaining suitable for use in continuum simulations.

This perspective allows the continuum model to serve not only as a numerical tool, but also as a bridge between microscopic physical processes and macroscopic device behavior.

In particular, this framework enables the incorporation of physically informed or data-driven constitutive relations, depending on the level of detail available for the underlying material.

3 Problem Statement and Motivation

Consider a conductive element with length L, cross-sectional area A, and resistivity ρ. Its electrical resistance is given by

$$R = \frac{\rho L}{A}. \tag{1}$$

Under mechanical loading, both the geometry and the intrinsic material properties of the conductor may change. In particular, the length, cross-sectional area, and resistivity become functions of deformation, making the resistance dependent on the mechanical state.

Using Ohm's law,

$$R = \frac{V}{I}, \tag{2}$$

and expressing voltage and current in terms of field quantities,

$$V = EL, \qquad I = JA, \tag{3}$$

we obtain

$$\frac{EL}{JA} = \frac{\rho L}{A}. \tag{4}$$

Canceling the geometric factors L and A leads to the local relation

$$\mathbf{J} = \sigma \mathbf{E}, \qquad \sigma = \frac{1}{\rho}, \tag{5}$$

where $\mathbf{E} = -\nabla V$ is the electric field. This formulation expresses Ohm's law in a local (continuum) form, independent of the specific geometry and applicable to spatially varying fields.

Combining this with charge conservation in the steady state,

$$\nabla \cdot \mathbf{J} = 0, \tag{6}$$

yields the conductivity equation

$$\nabla \cdot (\sigma \nabla V) = 0. \tag{7}$$

This equation provides a continuum description of electrical conduction in materials, where the conductivity σ may vary spatially. It belongs to the class of elliptic partial differential equations in divergence form [2].

In piezoresistive materials, the key feature is that the conductivity is no longer constant, but depends on the mechanical deformation of the material. At a macroscopic level, this dependence can be expressed as

$$\sigma = \sigma(\varepsilon(u)), \tag{8}$$

where the strain tensor is given by[1]

$$\varepsilon(u) = \frac{1}{2}(\nabla u + \nabla u^T). \tag{9}$$

This relation can be interpreted as an effective description of how mechanical deformation influences electrical transport. From a physical perspective, the dependence of conductivity on strain originates from strain-induced modifications of the electronic structure and carrier transport properties, which are affected by lattice deformation [3].

The resulting model therefore introduces a nonlinear coupling between mechanical deformation and electrical conduction: the mechanical field u determines the strain $\varepsilon(u)$, which modifies the conductivity σ, and this in turn affects the electrical potential V through the conductivity equation.

This illustrates that both geometric deformation and intrinsic material changes contribute to the variation of resistance, providing the basis for modeling the piezoresistive effect within a continuum framework.

[1] The strain tensor is defined as the symmetric part of the displacement gradient, which removes rigid body rotations and captures the true deformation of the material.

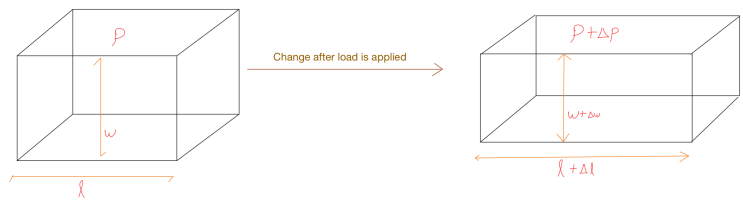

Figure 2: Change of geometric and material properties under mechanical deformation. L denotes the length of the conductor, W its characteristic width, and $A = W^2$ the cross-sectional area (assuming a square cross-section). The material property ρ represents the electrical resistivity, which may vary under mechanical deformation. Both geometric changes and intrinsic material variations contribute to the overall change in resistance.

4 Mathematical Model

Based on the physical considerations discussed above, the electromechanical behavior of the material is described at the continuum level by a coupled system of partial differential equations.

The mechanical response is governed by the equations of elasticity,

$$\nabla \cdot \boldsymbol{\sigma}(u) = 0, \tag{10}$$

where u denotes the displacement field and $\boldsymbol{\sigma}(u)$ is the stress tensor, which depends on the strain through a constitutive relation.

The electrical response is described by the conductivity equation,

$$\nabla \cdot (\sigma(\varepsilon(u))\nabla V) = 0, \tag{11}$$

where V is the electrical potential and the conductivity σ depends on the strain tensor $\varepsilon(u)$.

Together, these equations define a coupled system in which the mechanical field u determines the strain $\varepsilon(u)$, which in turn modifies the conductivity σ, and thereby influences the electrical potential V.

From a physical perspective, this coupling provides an effective continuum description of how mechanical deformation alters the underlying electronic transport properties of the material.

The system is nonlinear due to the dependence of the conductivity on the strain field.

This formulation serves as the basis for the computational modeling of the piezoresistive effect, linking continuum mechanics with strain-dependent electrical conduction.

5 State of Research

The piezoresistive effect has been extensively investigated in both experimental and engineering contexts, particularly in semiconductor-based sensors and microelectromechanical systems (MEMS) [4, 1]. Early studies established the strong dependence of electrical conductivity on mechanical strain in materials such as silicon, highlighting the importance of crystal structure and carrier transport in determining the sensitivity of the effect.

From a materials physics perspective, piezoresistivity originates from strain-induced modifications of the electronic structure and charge transport properties. Mechanical deformation can

alter band structure, effective masses, and scattering mechanisms, which in turn influence carrier mobility and electrical conductivity [3]. These effects are well understood at the microscopic and transport levels.

In practical modeling and engineering applications, however, this coupling is typically introduced in a simplified manner, where the conductivity is treated as an empirical function of strain. While such approaches are effective for device design, they do not provide a transparent link between microscopic transport mechanisms and macroscopic behavior.

At the continuum level, partial differential equation models are widely used to describe both elasticity and electrical conduction. Nevertheless, the incorporation of strain-dependent transport properties into these models is usually carried out phenomenologically, without a systematic connection to the underlying physics.

This reveals a gap between physically grounded transport descriptions and continuum-level modeling. Bridging this gap in a computationally consistent way remains an open and relevant problem, particularly for materials and devices where electromechanical coupling plays a central role.

Based on these considerations, the proposed study is guided by the following research questions:

- How can strain-dependent conductivity be incorporated into a consistent continuum model of electromechanical coupling?
- How does mechanical deformation influence the spatial distribution of electrical potential in conductive materials?
- What effects arise from the nonlinear coupling between mechanics and electrical transport?
- To what extent can flexible (e.g. data-driven or physics-informed) approaches improve the modeling of $\sigma(\varepsilon)$?

To address these questions, the study is structured around the following components:

- Physical background and modeling assumptions
- Formulation of the coupled electromechanical model
- Analysis of the nonlinear system
- Computational methods and numerical experiments
- Interpretation of results in the context of material behavior

6 Methodology

The proposed study follows a computational approach based on a continuum description of the coupled electromechanical system. The methodology is structured in several stages, combining physical modeling with numerical approximation.

First, the governing equations of elasticity and electrical conduction are formulated within a consistent continuum framework. The mechanical problem provides the displacement field u, from which the strain tensor $\varepsilon(u)$ is computed. This strain field enters the electrical problem through the strain-dependent conductivity $\sigma(\varepsilon)$.

As a baseline, the electrical conduction problem with constant conductivity is considered and solved numerically. This case admits an analytical solution and serves to validate the numerical

implementation. A finite element discretization is employed, and the numerical solution is compared against the exact solution to ensure correctness.

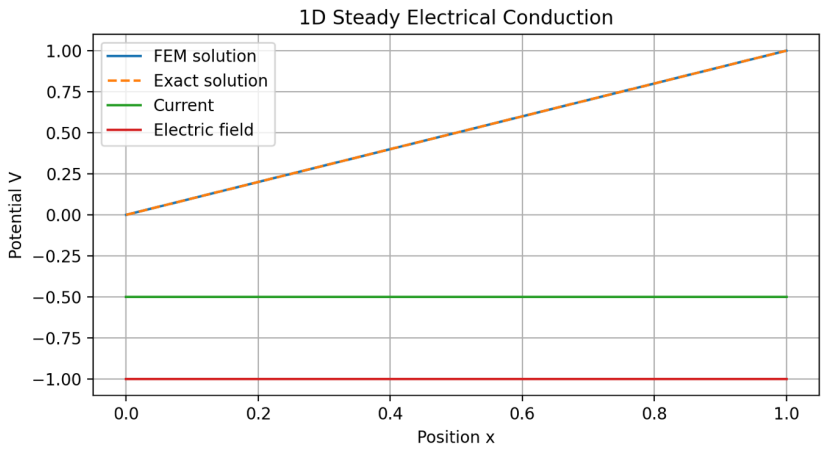

Figure 3: Finite element solution of the 1D steady conduction problem with constant conductivity compared with the analytical solution. The linear potential profile and constant electric field confirm the expected physical behavior.

This baseline case also provides a reference for interpreting deviations introduced by strain-dependent conductivity in the coupled problem.

Building on this, the full coupled system is addressed by introducing a spatially varying conductivity $\sigma(\varepsilon(u))$. The problem is solved iteratively: starting from an initial displacement field, the strain is computed, the conductivity is updated, and the electrical problem is solved. This procedure is repeated until convergence is achieved.

The resulting system is nonlinear due to the dependence of σ on $\varepsilon(u)$, and particular attention is given to the stability and convergence of the numerical scheme, as well as to the sensitivity of the solution with respect to deformation.

From a computational perspective, the finite element method (FEM) is used as the primary discretization technique due to its flexibility in handling heterogeneous material properties and complex geometries. The implementation emphasizes clarity and modularity, allowing systematic exploration of different forms of the strain–conductivity relation.

Beyond purely phenomenological models, the framework is designed to accommodate physically motivated constitutive relations. In particular, the function $\sigma(\varepsilon)$ can be informed by transport models derived from electronic structure calculations, or approximated using data-driven approaches when such data is available.

Overall, the methodology provides a transparent and extensible computational pipeline that connects continuum modeling with physically grounded descriptions of strain-dependent conductivity.

7 Expected Outcome

The expected outcome of this study is the development of a computational framework for analyzing strain-dependent electrical conduction in deformable materials. The framework combines continuum mechanics and electrical transport into a unified numerical model, enabling the systematic investigation of electromechanical coupling.

At the modeling level, the work provides a consistent formulation of conductivity as a strain-dependent quantity, together with a numerical implementation for solving the resulting coupled system. This allows for the analysis of how mechanical deformation influences electrical potential distribution, current density, and overall system sensitivity.

From a computational perspective, the study delivers a flexible pipeline that can accommodate different forms of the strain–conductivity relation, ranging from phenomenological descriptions to physically informed or data-driven models. This makes the framework adaptable to different materials and configurations.

Beyond the specific application, the results contribute to the understanding of nonlinear multiphysics systems involving coupling between deformation and transport phenomena. Potential applications include piezoresistive sensing, structural health monitoring, and functional materials design.

The developed framework also provides a foundation for future extensions toward multiscale modeling and more detailed transport descriptions.

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